

Project Report On

Link in Bio Tool

Developed By

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Submitted To



Geetanjali College of Computer Science and Commerce (BBA)
Saurashtra University Rajkot
2023-2024

Project Guide

Prof. Pranav Trivedi

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Acknowledgment

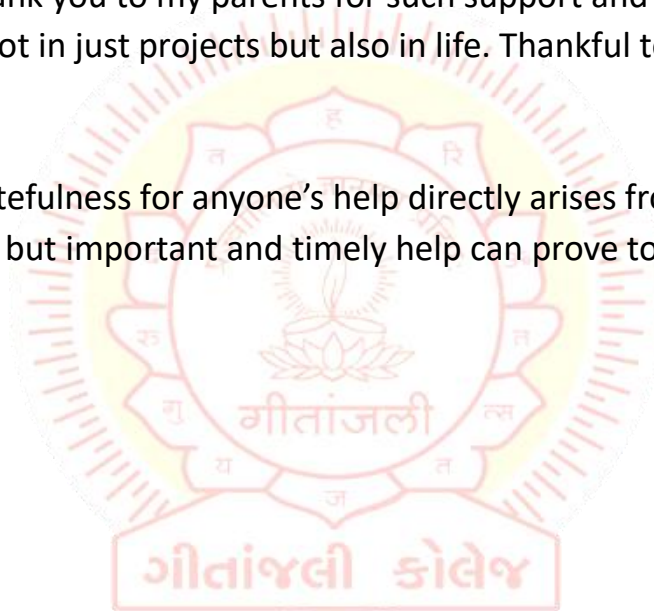
I am happy to submit my idea for the "Linkmonks (Link in bio tool)" Application to Saurashtra University, Rajkot for a BCA degree in the computer branch.

I am grateful to Prof. Brijesh Shah Head of the Department and all the faculty members of the Department of Computer Science for their kind support throughout this journey.

I take the privilege to acknowledge the elite authors of numerous books and papers and blogs we have referred to during this progress.

I also say a big thank you to my parents for such support and without them, I can do nothing, not in just projects but also in life. Thankful to my family for their support.

The feeling of gratefulness for anyone's help directly arises from the bottom of my heart. A small but important and timely help can prove to be a milestone in one's life.



Preface

In 21st Century of Technology, Computer and related service is the most useful, automatic and highly speed of work. B.C.A is course in which practical aspect is important as theoretical aspect. In the present world importance of education is increasing.

The project training in a student's life is like a live experience in the industry. If we have experience then & then we get top of the level. Most Important Factor of B.C.A Course is Project Work.

The Project Work is Must for Every Student in Different Government or Private Organization during the Study Period. We are developing project in 5th Semester of B.C.A we are the student of Geetanjali Group of Colleges Is Affiliated with Saurashtra University, Rajkot.

I created this project dynamic and in this project we provide select, insert, update and delete operation on client user can easily show site and give feedback. In client side there is so many features like view all links, delete & edit link if the user id and link user id is same, user profile, user posts count, specific user post, etc.

As this is fully and educative project site so it has made many limitations as per the fully furnished site view.

I have gone through and study necessary information, which is collected and presented in this report. I am trying my best in this project. I had taken almost care that the information provided in the project is true and project is in a smooth running condition and error free.

Project Profile

Project Title : Linkmonks

Development Software : VS Code

Front End : Reactjs, Tailwindcss, Shadcn-ui

Backend : Appwrite

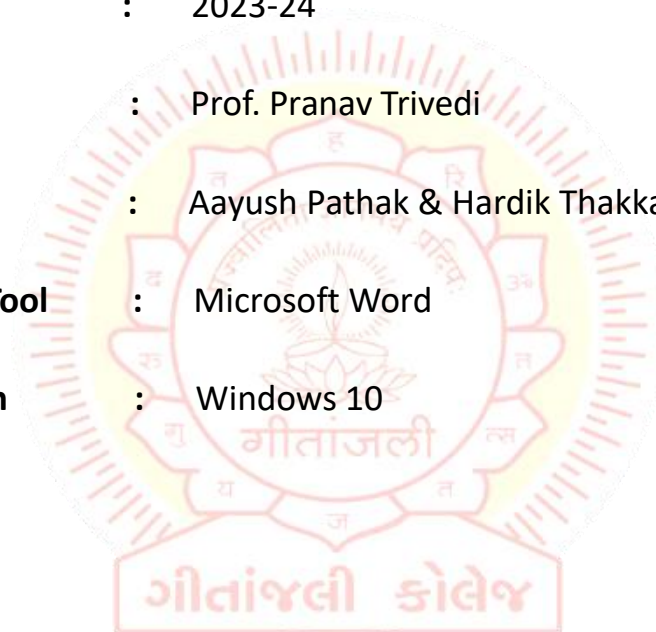
Year : 2023-24

Project Guide : Prof. Pranav Trivedi

Submitted By : Aayush Pathak & Hardik Thakkar

Documentation Tool : Microsoft Word

Operating System : Windows 10



Project Requirements

Processor	Core2Duo or above
Ram	Minimum 2GB Ram
Operating System	Any
Graphics Card	You Don't Need Graphic Card
Database	Appwrite Database
Language	React js



Software Development Life Cycle (SDLC)

Software Development Life Cycle Steps:

Here, this website was developed based on SDLC (Software Development Life Cycle). SDLC has a different step of development of websites or any type of application.

Following the SDLC steps are used in the website development: -

- System Requirement Analysis.
- Feasibility study
- System Design
- Code Generation
- Testing
- Implementation
- Maintenance

System Requirement Analysis:

In this phase, the development team visits the need for possible software requirements in micro-depth to develop the system.

The basic clearance of the website they informed by the system requirement analysis investigation team. Understanding fully what a project will deliver is critical to its success.

Feasibility study:

Feasibility means the analysis team's report first study and that after the decision to create this website.

This team internally shows the company's possibility of developing this type of website.

Tools and Technology, Economically, Government Rules & regulations are considered.

Based on the feasibility study the project 'Logistic Application' is feasible.

System Design:

This phase works on the creation of the first pre-model of the whole system.

In terms of the end user/server technology, the number of tiers needed for the package architecture, the database design, the data structure design, etc., are all defined in this phase.

Implementation:

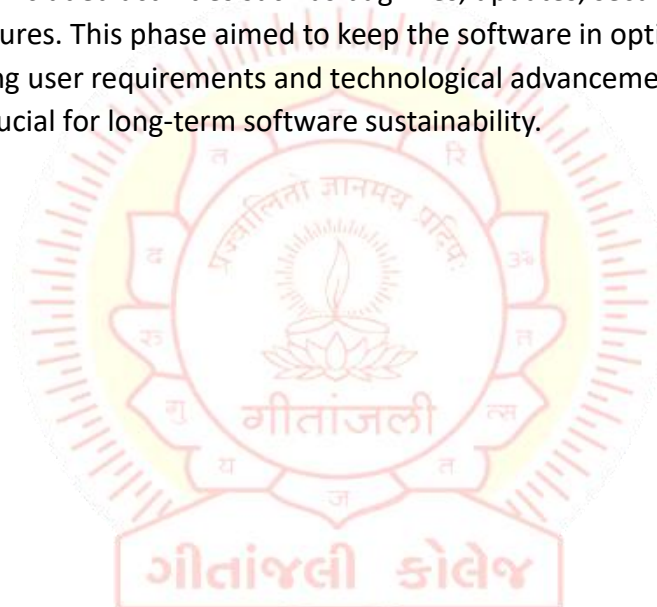
Implementation was the phase where the actual coding and development took place. Programmers and developers wrote the software code based on the design specifications. This phase included rigorous testing to identify and fix any coding errors or bugs.

Testing:

The testing phase involved systematically evaluating the software to ensure it met quality standards. Different testing levels, such as unit testing, integration testing, and user acceptance testing, were conducted to validate the software's functionality, performance, and reliability. Defects were identified and addressed during this phase.

Maintenance:

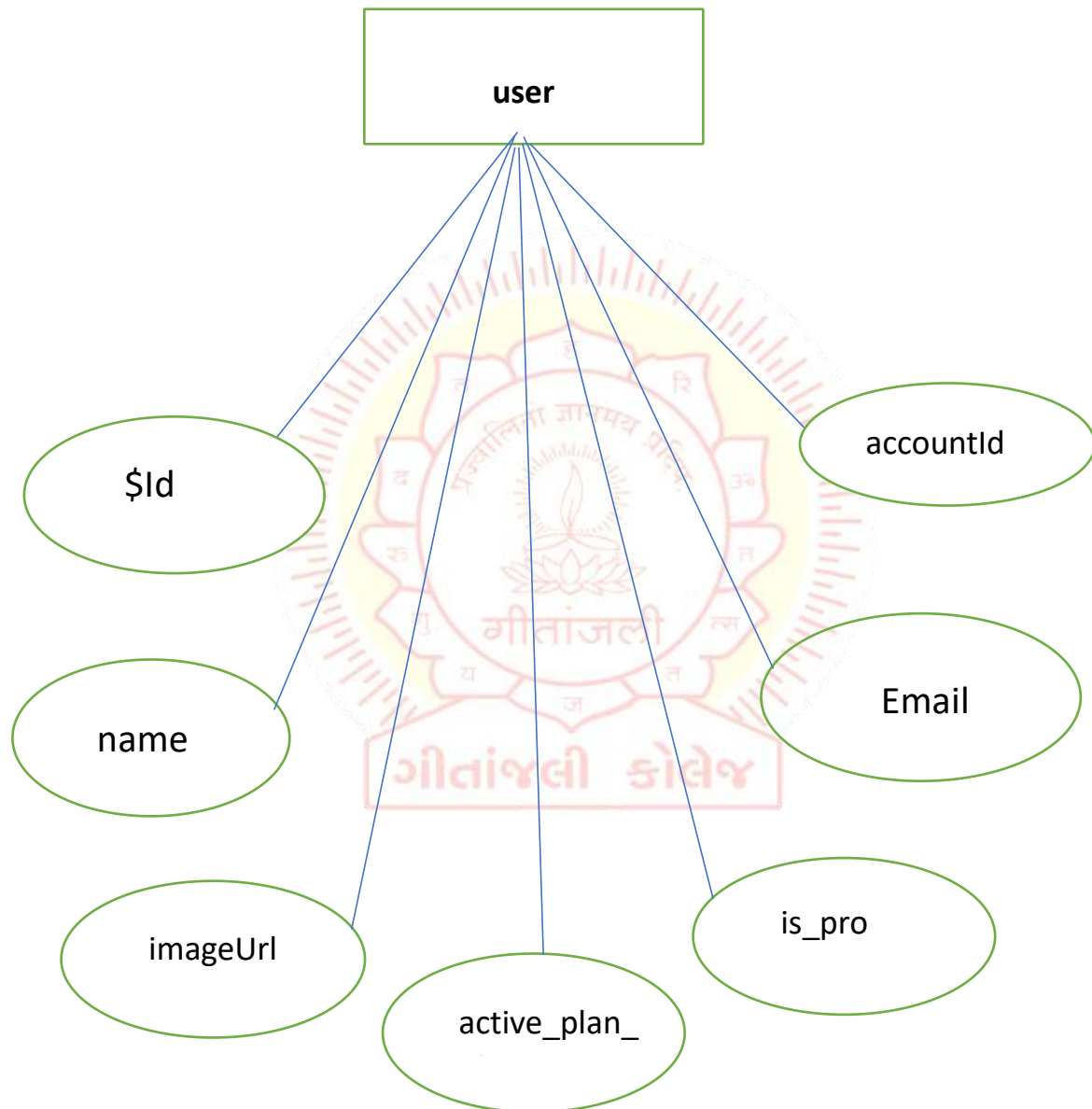
The maintenance phase represented the ongoing support and enhancement of the software post-deployment. It included activities such as bug fixes, updates, security patches, and the addition of new features. This phase aimed to keep the software in optimal condition and aligned with changing user requirements and technological advancements. Effective maintenance was crucial for long-term software sustainability.



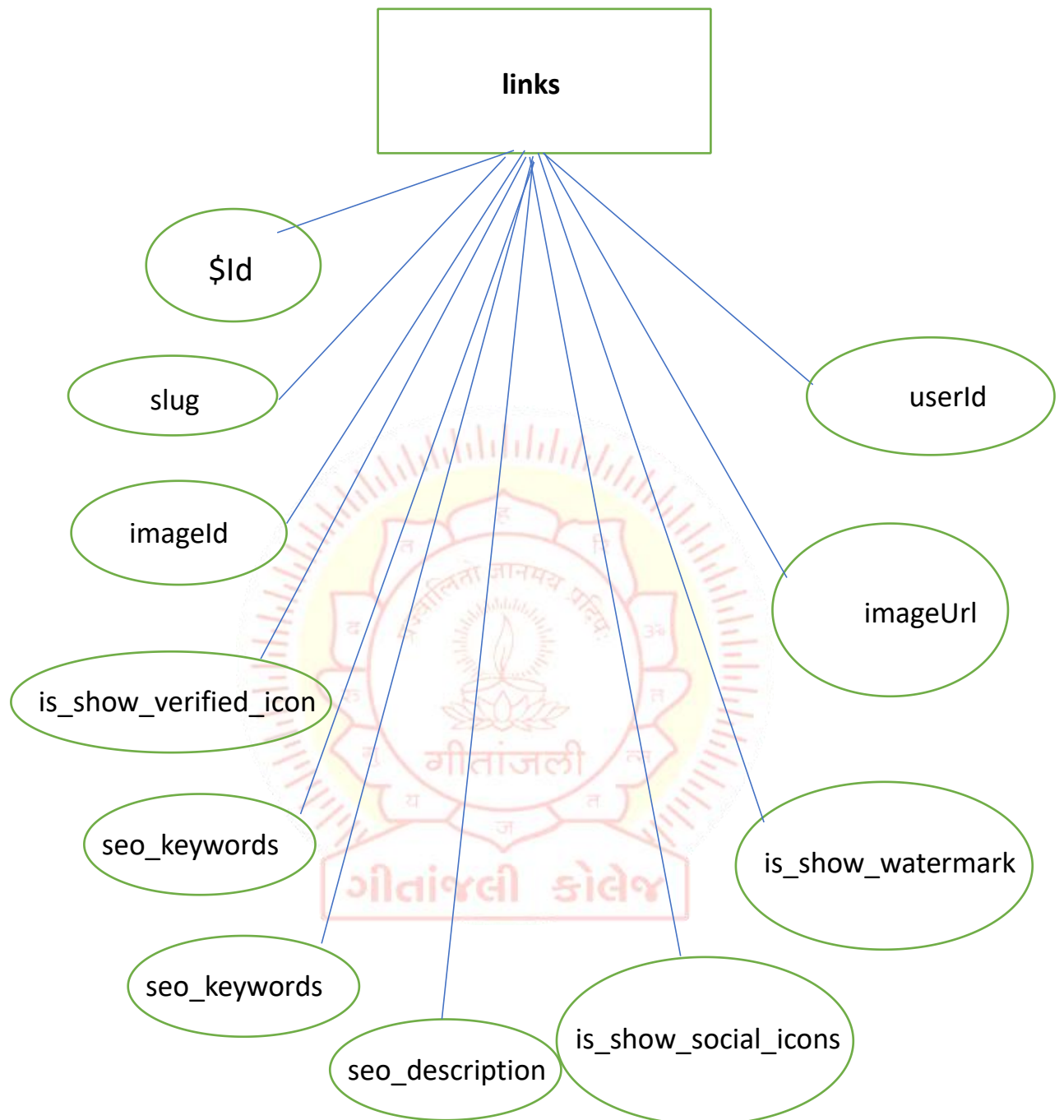
ER-Diagram

An entity Relationship Diagram (ERD) is a data modeling technique that graphically illustrates an information system's entities and the relationship between those entities. An ERD is a conceptual and representation model of data used to represent the entity from work infrastructure.

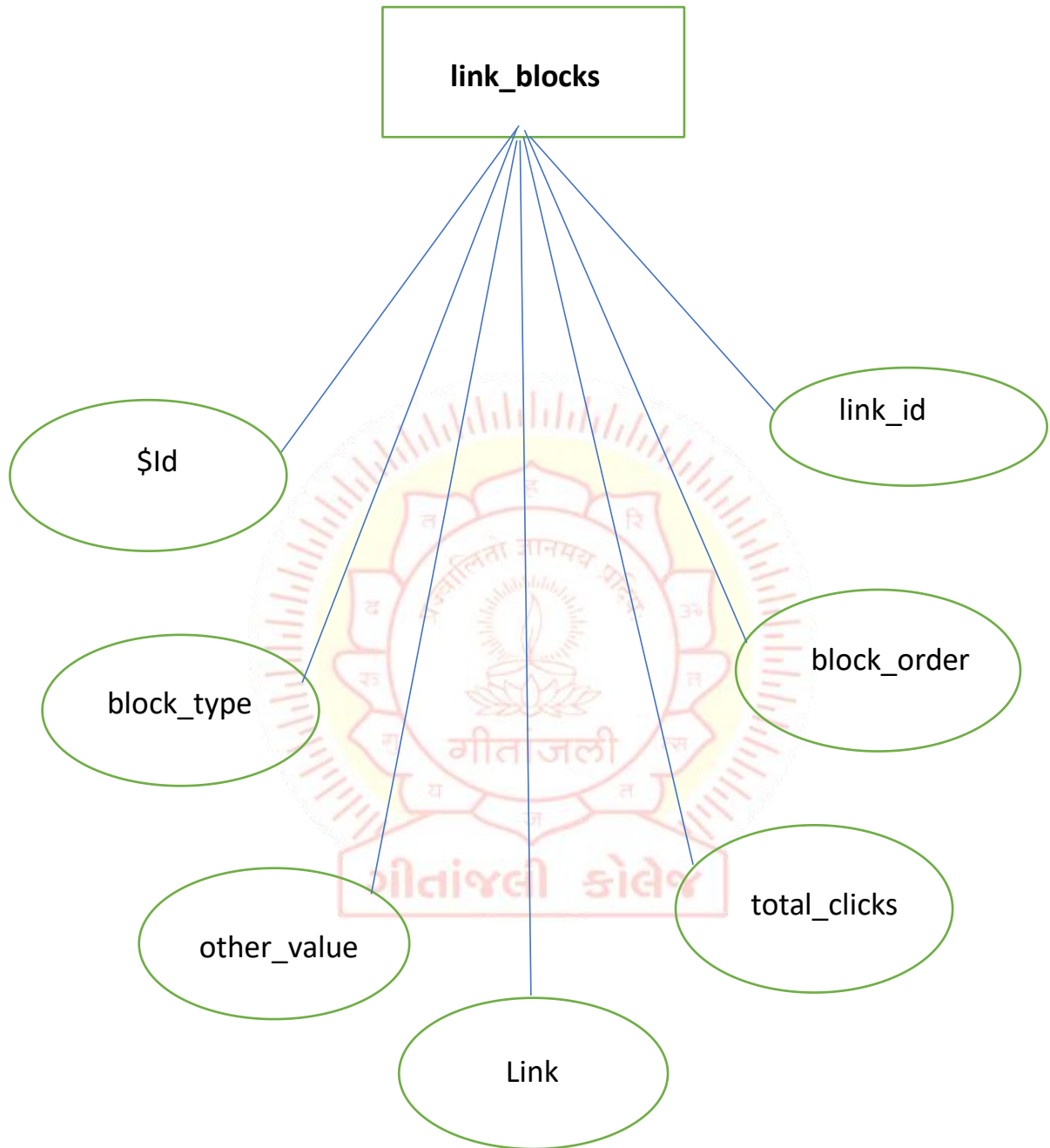
1) user



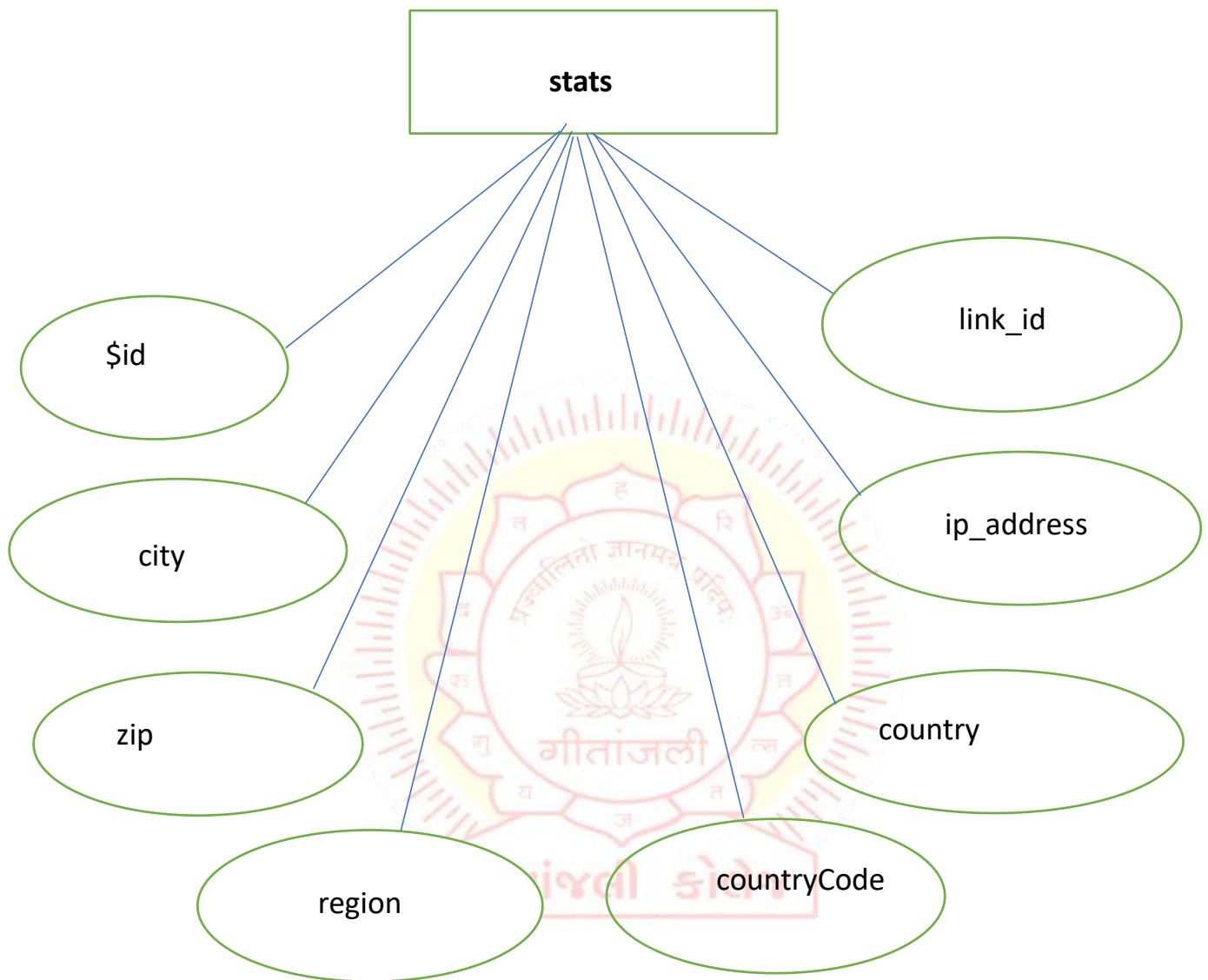
2) links



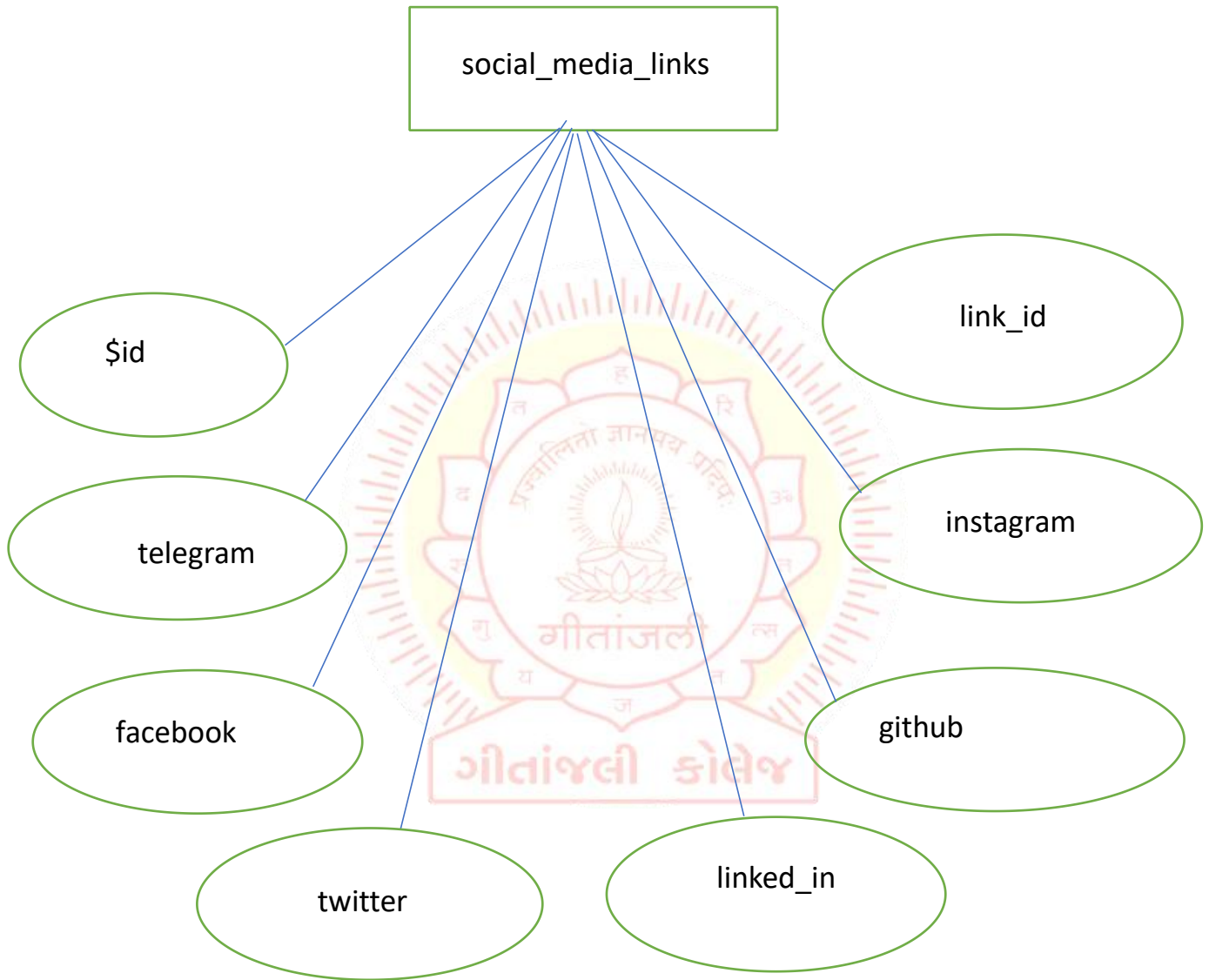
3) link_blocks



4) stats



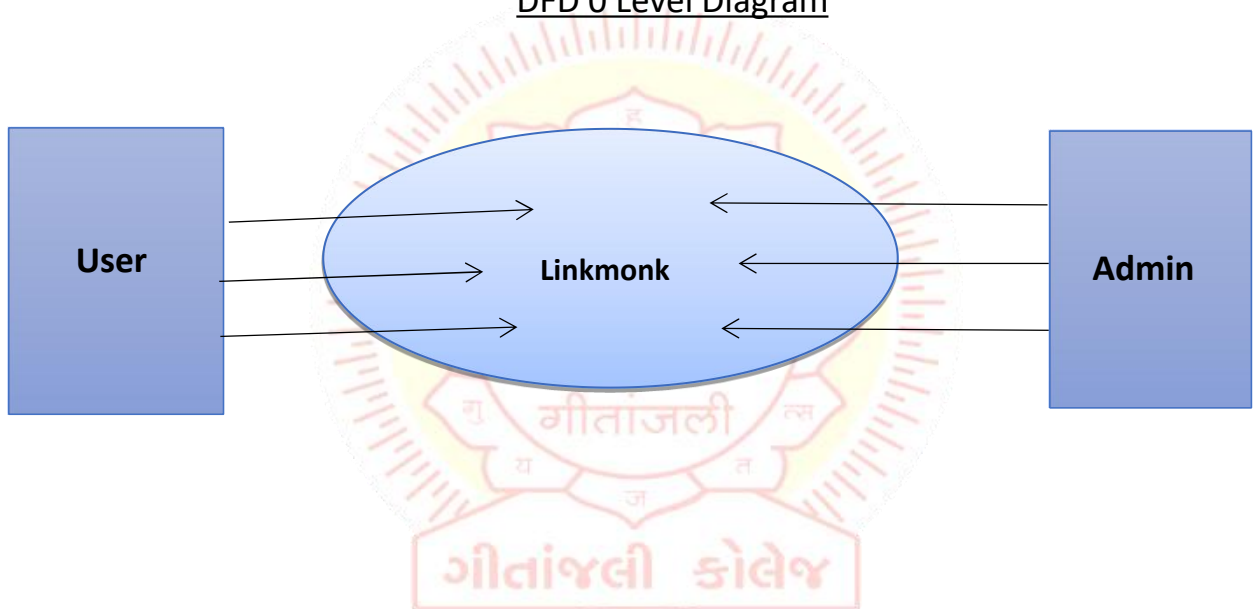
5) social_media_links



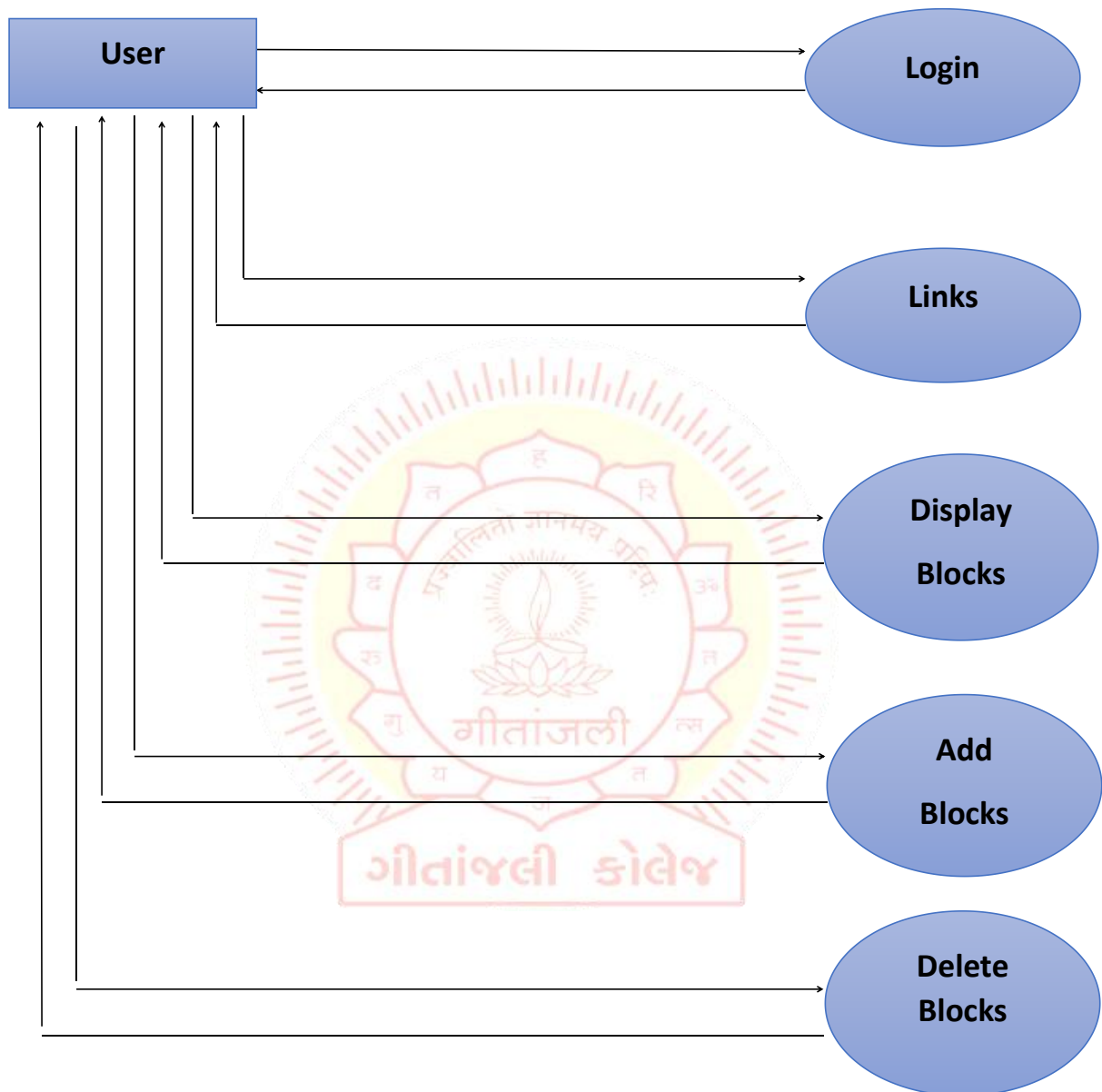
System Design Tools [DFD's]

DFD (Data Flow Diagram) :

DFD 0 Level Diagram



DFD 1 Level Diagram



Data Dictionary

➤ Database Collections :-

- 1) users
- 2) links
- 3) link_blocks
- 4) stats
- 5) social_media_links
- 6) plans



1) user

Sr No.	Fields	Data Type	Size	Constraints
1	\$id	string	-	Primary key
2	accountId	string	100	-
3	email	string	100	-
4	imageUrl	string	100	-
5	name	string	100	-
6	active_plan_id	string	25	-
7	is_pro	boolean	-	-

2) links :-

Sr No.	Fields	Data Type	Size	Constraints
1	\$id	string	-	Primary key
2	userId	string	25	-
3	slug	string	20	-
4	imageUrl	string	100	-
5	imageId	string	25	-
6	is_show_verified_icon	boolean	-	-
7	is_show_social_icons	boolean	-	-
8	is_show_watermark	boolean	-	-
9	seo_keywords	string	150	-
10	seo_description	string	150	-
11	seo_title	string	75	-
12	ga_tag	string	25	-
13	description	string	150	-
14	is_private	Boolean	-	-

3) link_blocks:-

Sr No.	Fields	Data Type	Size	Constraints
1	\$id	String	-	Primary key
2	link_id	String	25	-
3	block_type	String	25	-
4	other_values	String	500	-
5	Link	url	-	-
6	total_clicks	Integer	-	-
7	block_order	Integer	-	-
8	is_private	Boolean	-	-
9	restriction_type	String	10	-
10	scheduled_block_id	String	25	-
11	is_featured	Boolean	-	

4) stats:-

Sr No.	Fields	Data Type	Size	Constraints
1	\$id	String	-	Primary key
2	link_id	varchar	50	-
3	ip_address	Ip	-	-
4	city	String	10	-
5	zip	integer	-	-
6	country	String	60	-
7	region	String	60	-
8	countryCode	String	16	-
9	Referrer	Url	-	-
10	Total_views_by_ip	Integer	-	-

5) social media links:-

Sr No.	Fields	Data Type	Size	Constraints
1	\$id	String	-	Primary key
2	link_id	varchar	50	-
3	telegram	string	30	-
4	instagram	String	30	-
5	facebook	String	50	-
6	github	String	20	-
7	twitter	String	20	-
8	linked_in	String	100	-
9	snapchat	String	100	-
10	pinterest	String	100	-

6) plans:-

Sr No.	Fields	Data Type	Size	Constraints
1	\$id	String	-	Primary key
2	price	String	100	-
3	Plan_name	string	30	-
4	max_links_allowed	integer	-	-
5	max_blocks_per_link	integer	-	-
6	Visitor_limits	Integer	-	-
7	Social_media_links	Boolean	-	-
8	Google_analytics	Boolean	-	-
9	Seo_optimization	Boolean	-	-
10	Verified_badge	Boolean	-	-
11	Support_24/7	Boolean	-	-

12	Animations	Boolean	-	-
13	Custom_domain	Boolean	-	-
14	Sub_links	Boolean	-	-
15	Scheduled_blocks	Boolean	-	-
16	Email_subscription	Boolean	-	-
17	Advanced_statistics	Boolean	-	-
18	Multiple_tabs	Boolean	-	-
19	Plan_desc	String	150	-
20	Plan_type	String	25	-
21	Monthly_price	String	100	-
22	Yearly_price	String	100	-
23	Lifetime_price	String	100	-
24	payment_link	Link	-	-

Test Cases

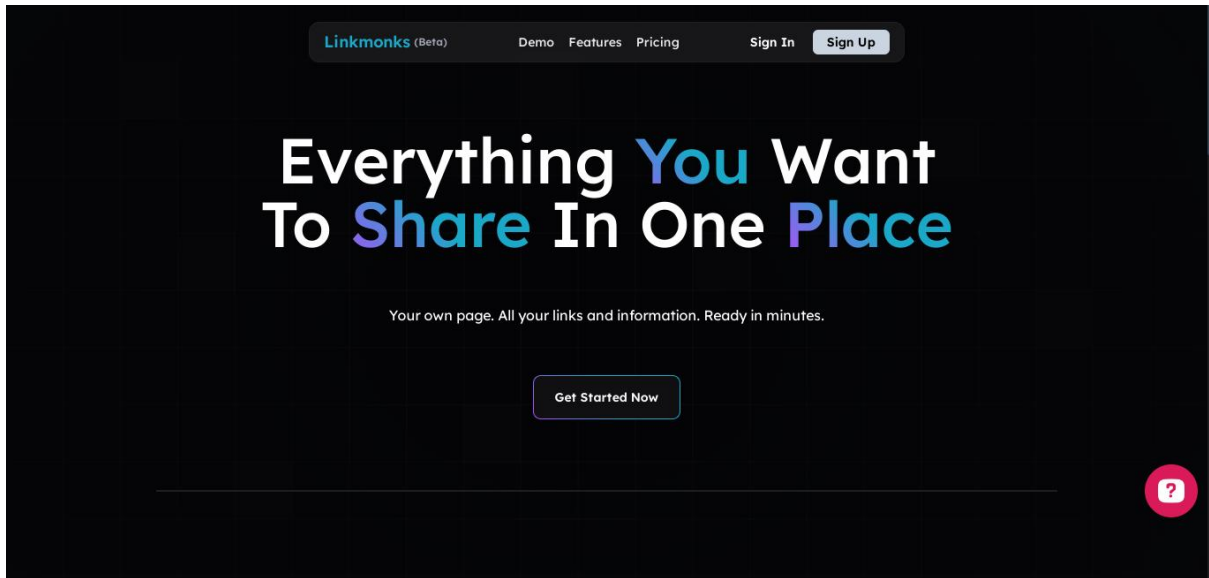
Case Name	Case Description	Actual Output	Test Result
Register	Register with new account	Register with the new account while already taken email cannot be registered again	Pass
Login	A window with a two textbox email and password.	Write correct email and password and login successfully.	Pass
Home	Screen Displays all menus and create link form .	Screen Displays all menu or name of the Linkmonks and create link form.	Pass
Single link	Display single link page with all the tabs of settings of link	Display single link page with all the tabs of settings of link toggle multiple links	Pass
Add Blocks	Add multiple blocks,reorder& edit it	Created new blocks, reordered & edited also it woks perfectly,	Pass
Remove Blocks	Remove blocks	Removed some blocks and updated the blocks and its working perfectly	Pass
Stats tab	Display stats with proper data	Displayed all the stats with proper data of visitors in the selected link	Pass
Add Restriction	Add restriction for the free plan users that it is not accessible by the free plan users	Added restriction with the only pro users can access stats tab	Pass

Update account	Updated name and password	Updated name and password by entering old and new password works perfectly	Pass
Update Social Media Link	Update social media links in the selected link	Updated social media links workins perfectly	Pass



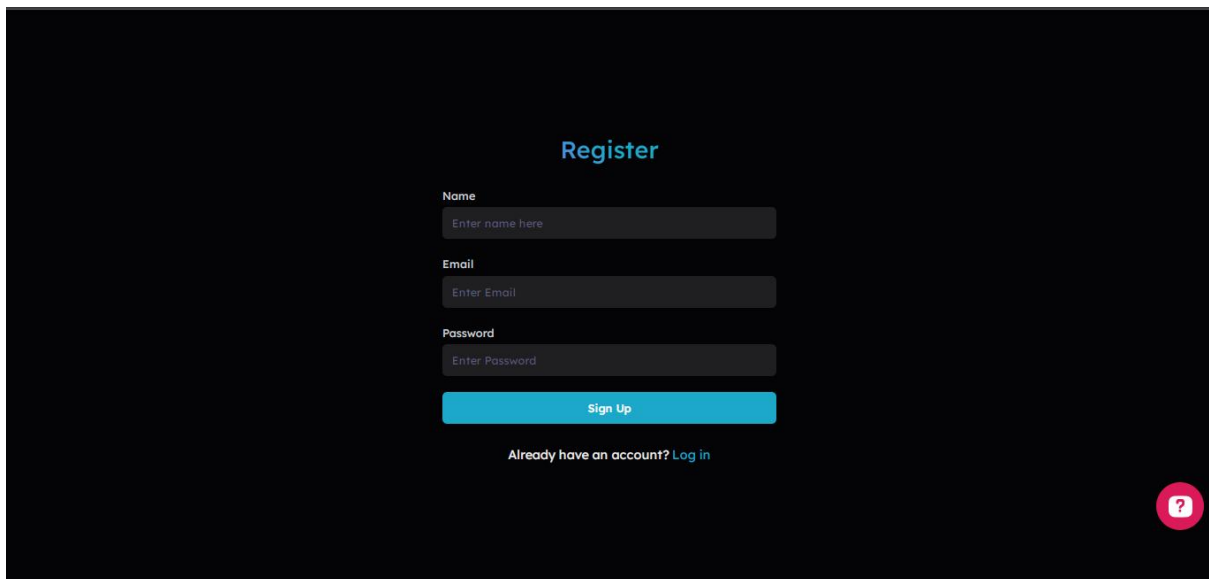
Screenshots

- **Landing Page:-**



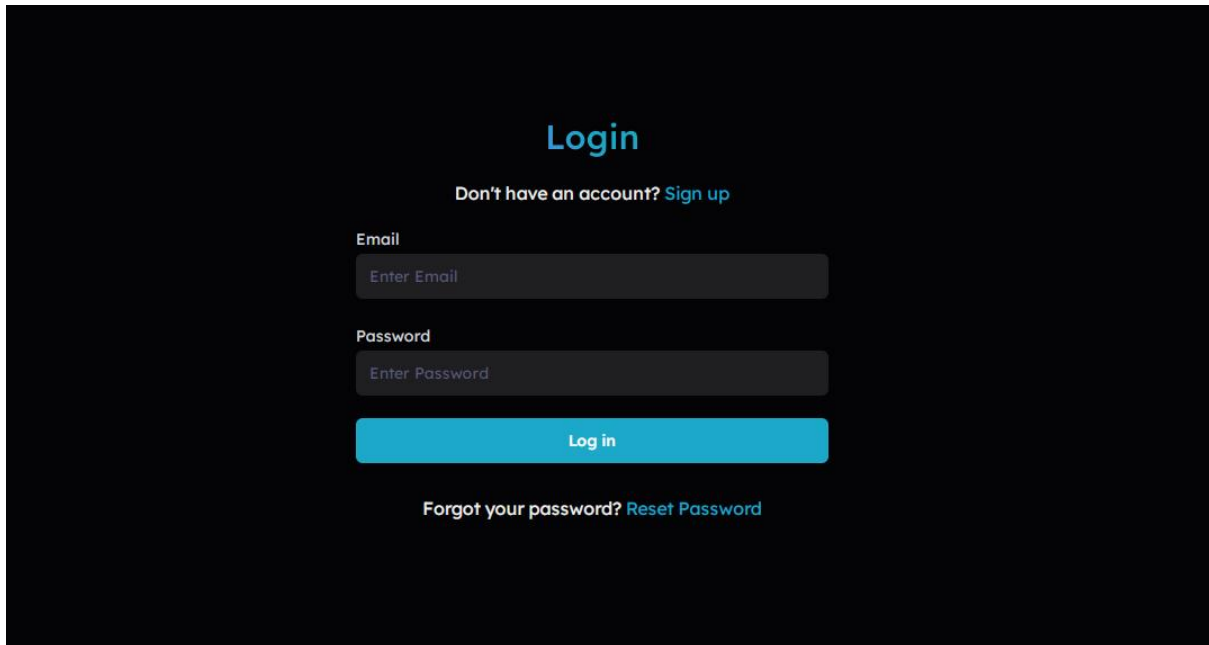
- This page is a basic landing page which is accessible by authorized and unauthorized users so this is our first page where we can see which features we provide and also which feature is available in pro and free plan using pricing cards

- **Register Page:-**



- This is a register page where users can register by entering their email and password and name. If the email is already taken then we display an error message that the email is already taken please try to login or enter new email. This page also contains the login page link If an user is already have an account then they can go to the login page by clicking on it.

- **Login Page:-**



Login

Don't have an account? [Sign up](#)

Email

Enter Email

Password

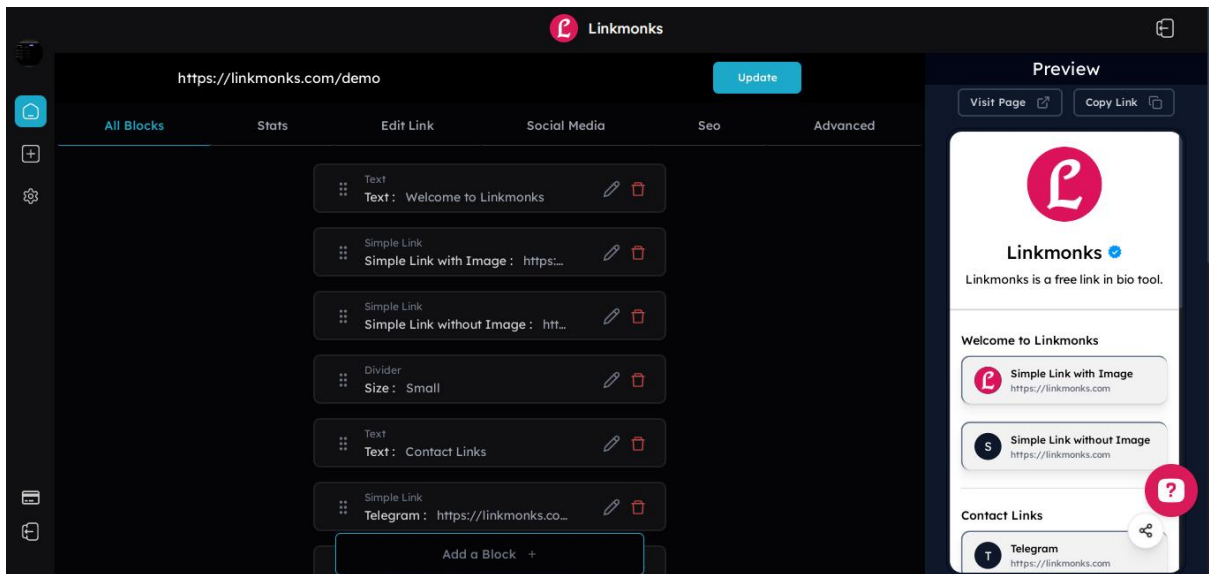
Enter Password

[Log in](#)

Forgot your password? [Reset Password](#)

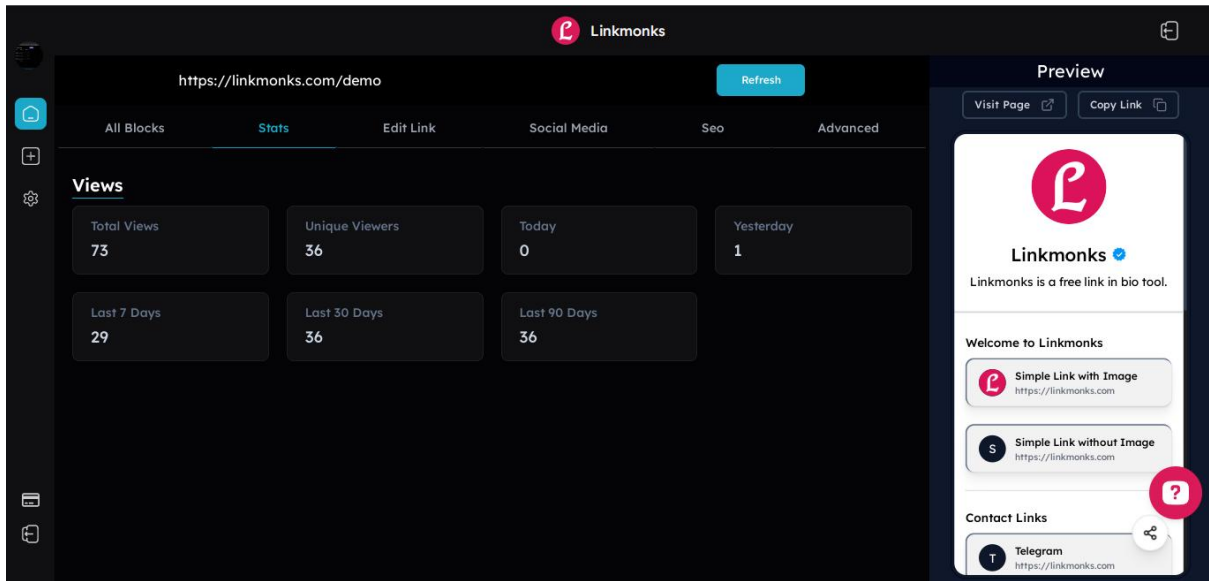
- This is a login page which is used to authenticate user to access their account in linkmonks. If an user is not having an account then the error. And if the password and email is entered correctly then user logs in to their account

- **HomePage:-**

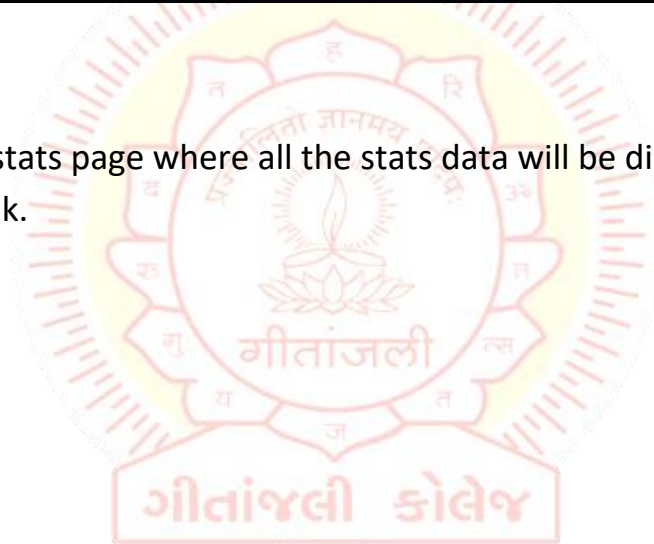


- This is home page where user can select links and add blocks in it. This page contains multiple tabs that user can manage their link in a single page by navigating through the multiple tabs

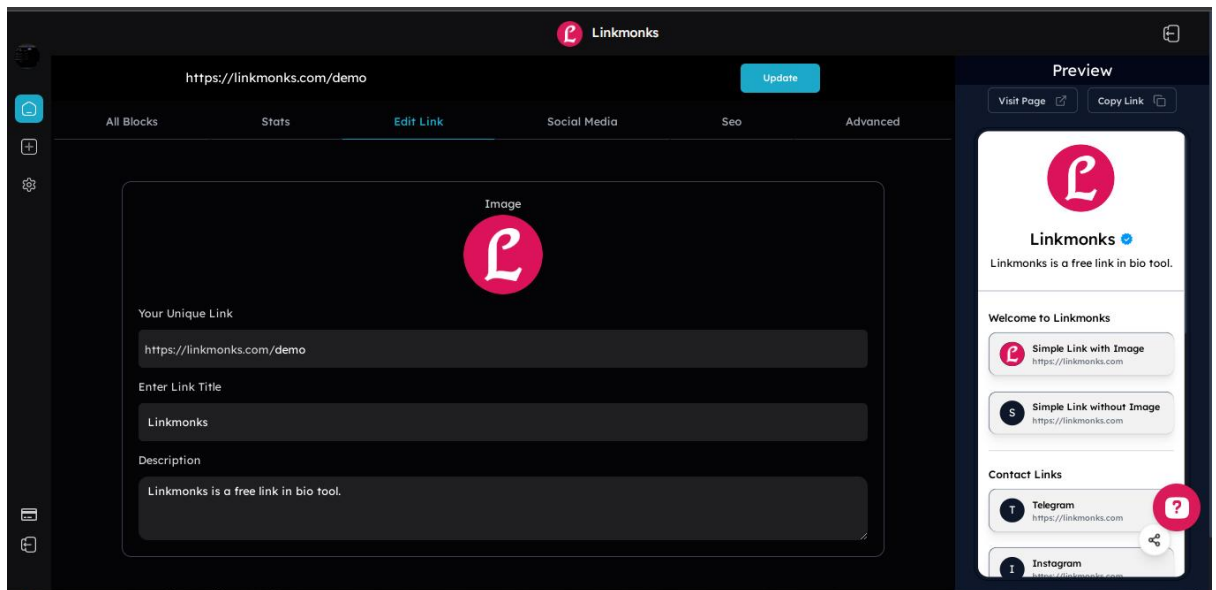
- **Stats Page:-**



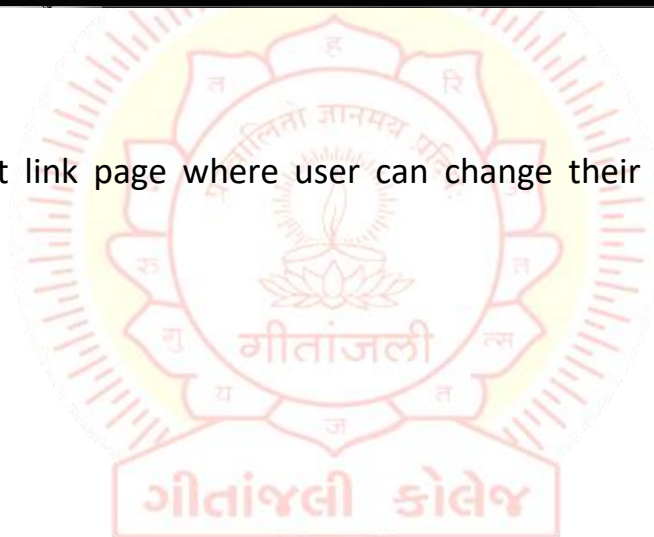
- This is the stats page where all the stats data will be displayed of the selected link.



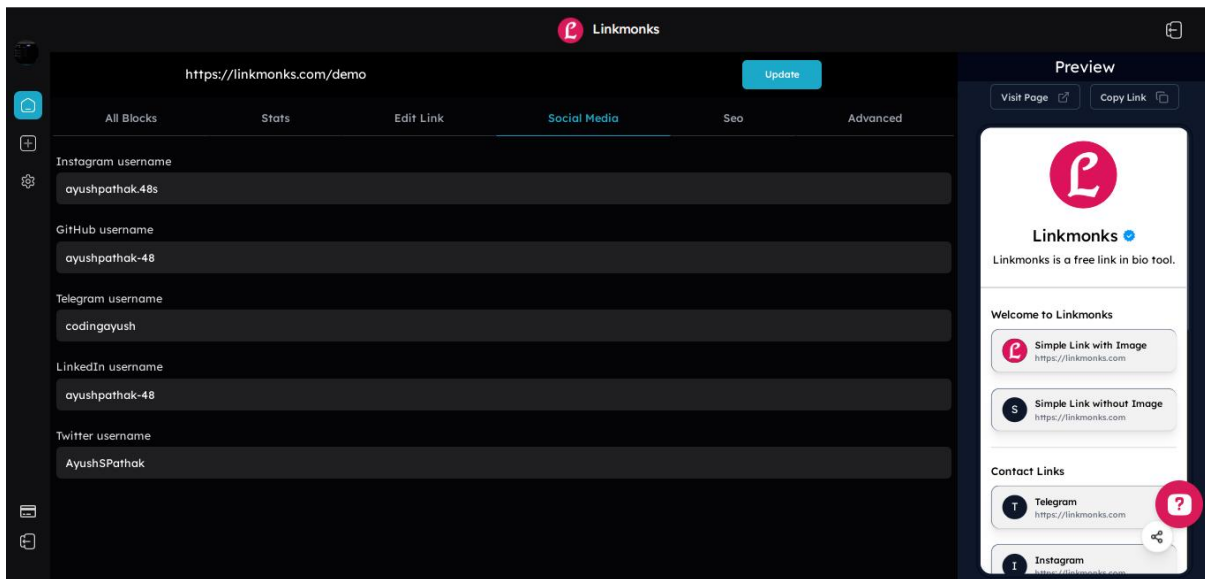
- **Edit Link:-**



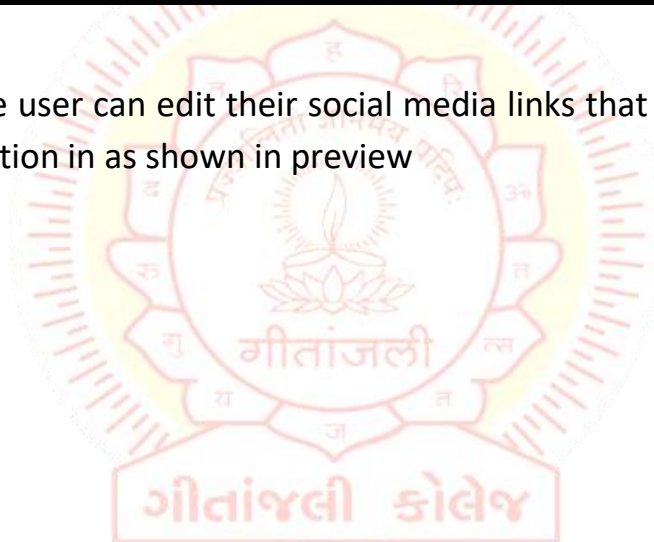
- This is an edit link page where user can change their slug and title and description



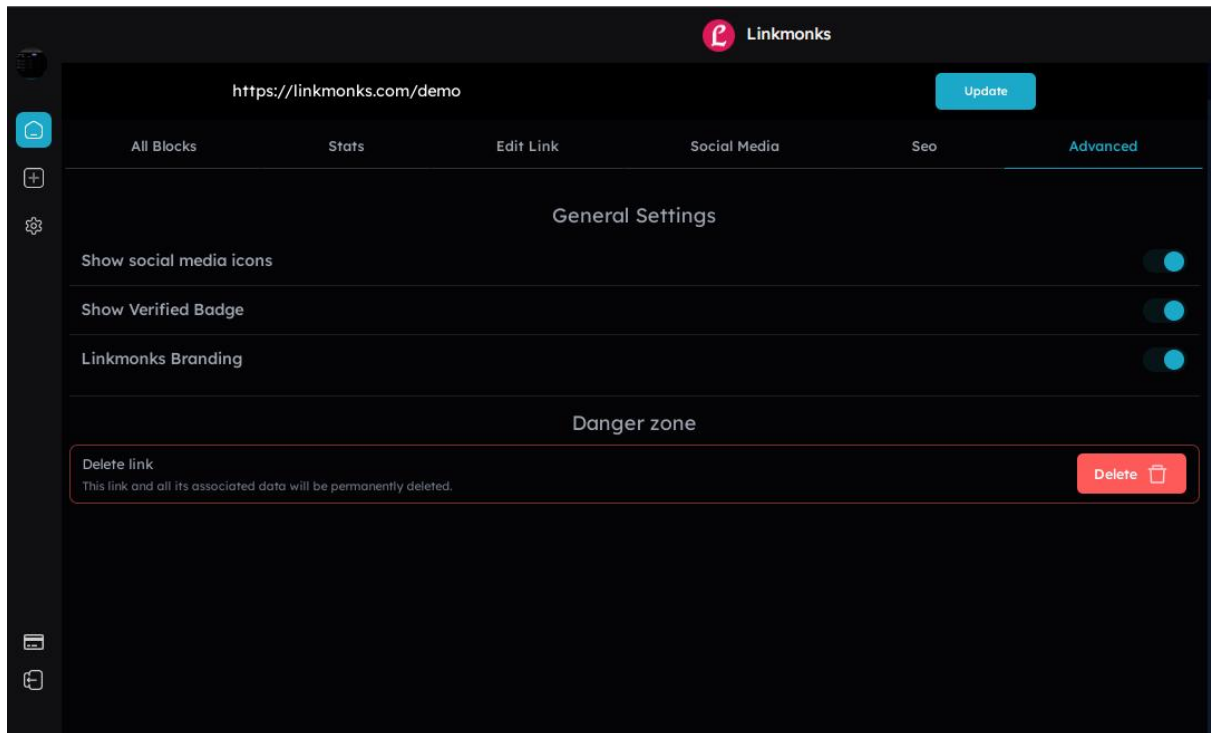
- update social media:-



- In this page user can edit their social media links that is displayed below the description in as shown in preview

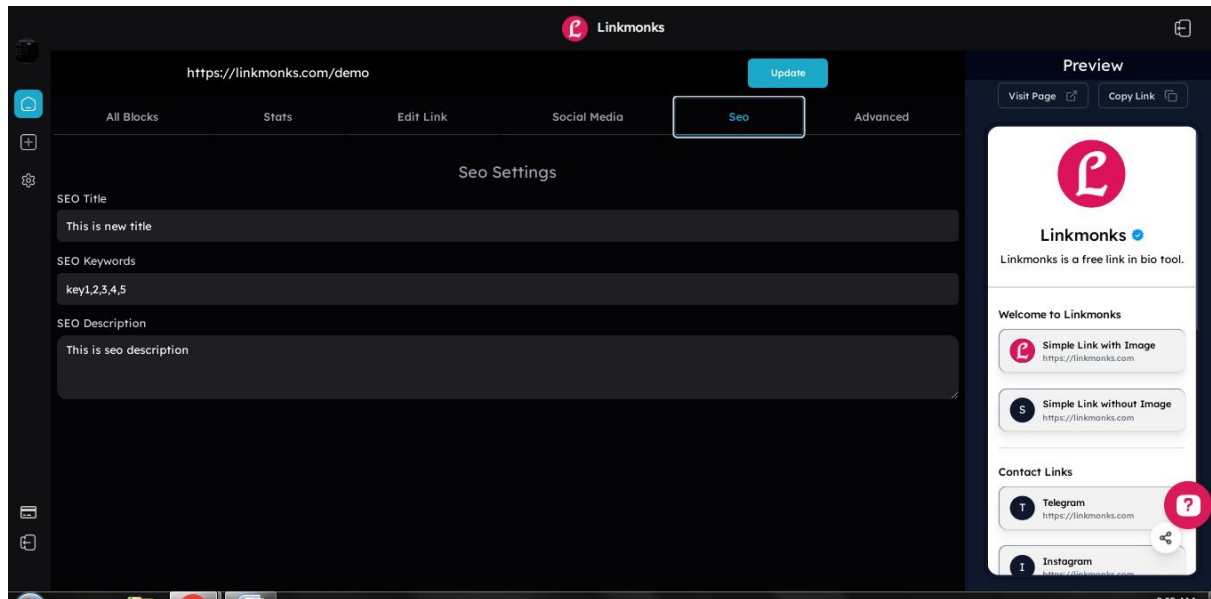


- **Advanced Link Settings:-**

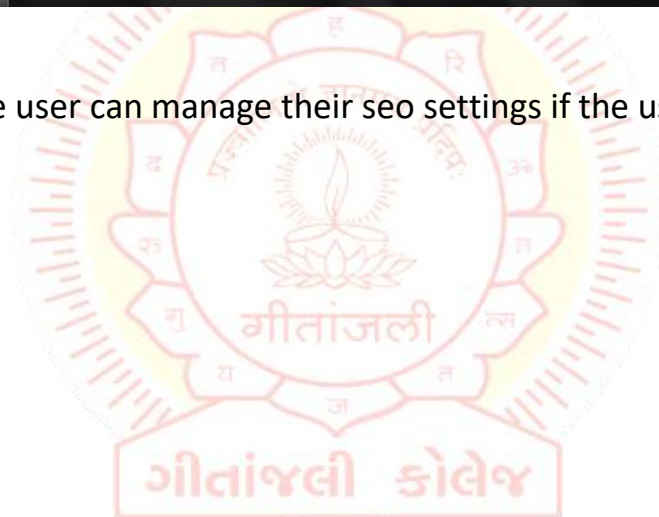


- This is an advanced link settings page where user can delete their link or toggle social media links whether to show this or not
- User can also enable verified badge if the user is a pro user where free user doesn't have access to the toggle button.

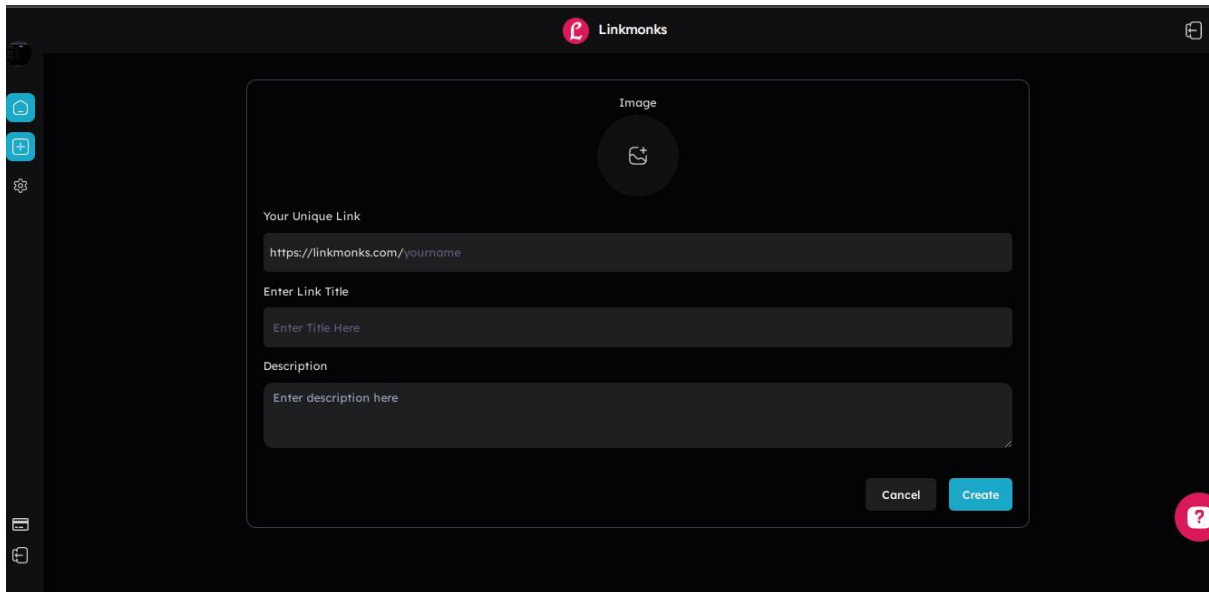
- **Seo Settings:-**



- In this page user can manage their seo settings if the user is a pro user



- **Create link:-**

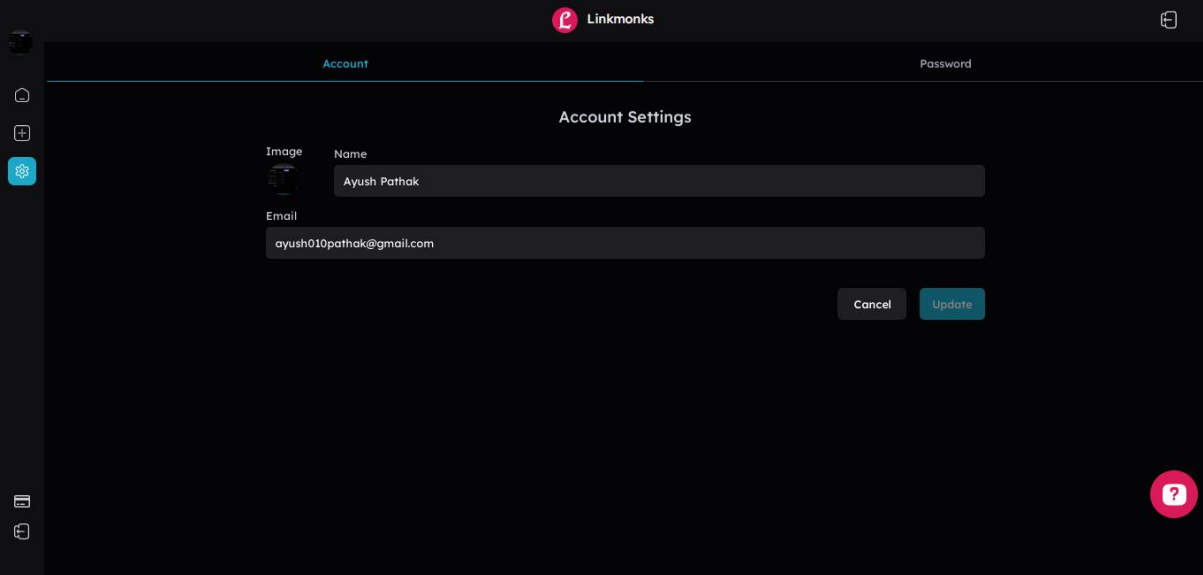


The screenshot shows the Linkmonks web interface for creating a new link. The form is titled 'Image' and includes a circular icon placeholder. Below this, there are three input fields: 'Your Unique Link' (pre-filled with 'https://linkmonks.com/yourname'), 'Enter Link Title' (placeholder: 'Enter Title Here'), and 'Description' (placeholder: 'Enter description here'). At the bottom right of the form are 'Cancel' and 'Create' buttons. The interface has a dark theme with a sidebar on the left and a top navigation bar.

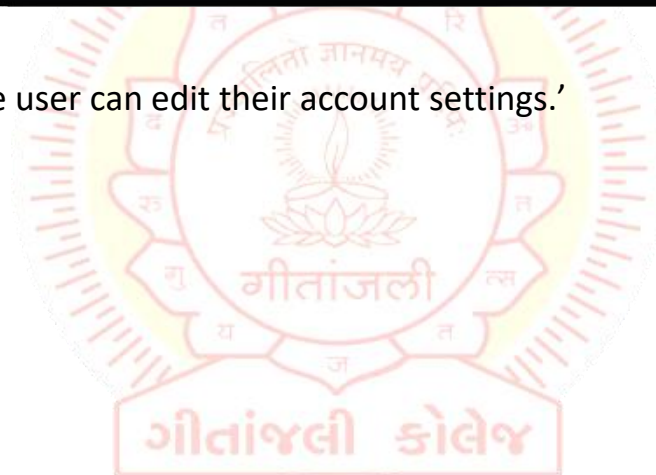
- This is a create link page where user can create links limited depends on their current plan.



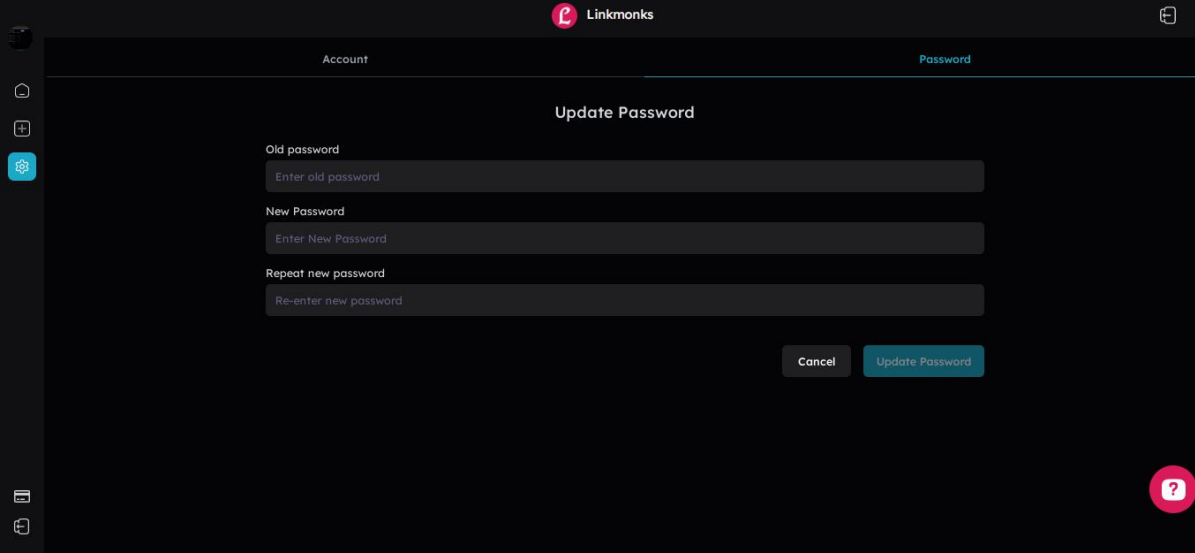
- **Account Settings :-**



➤ In this page user can edit their account settings.'



- **Update password:-**

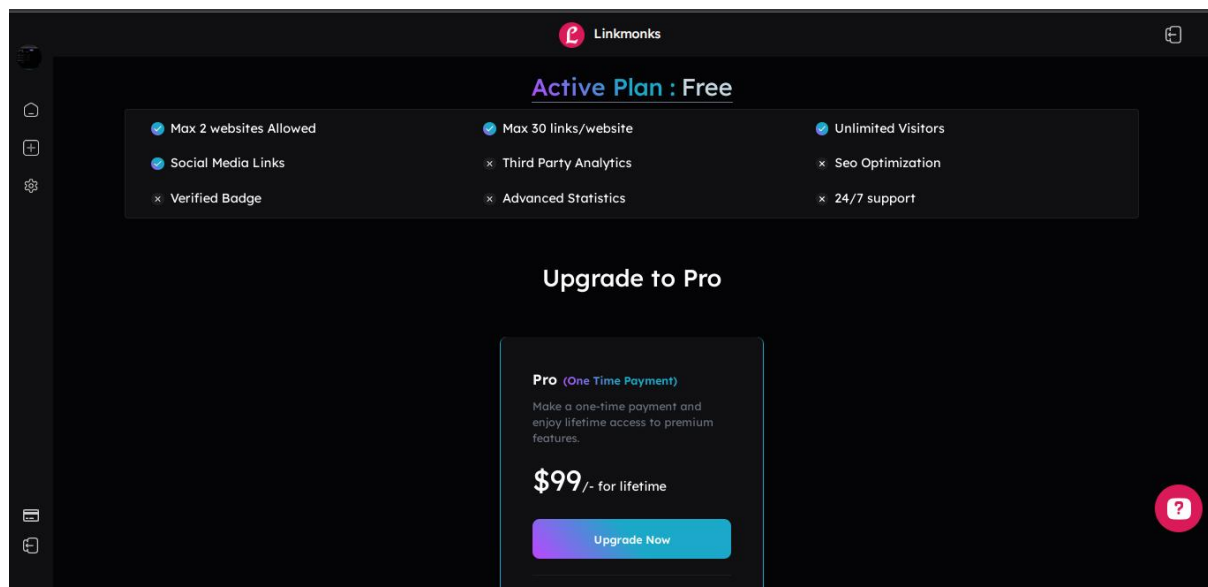


The screenshot shows a web interface for 'Linkmonks' with a dark theme. At the top, there's a navigation bar with 'Account' and 'Password' tabs. The 'Update Password' form is centered, featuring three input fields: 'Old password' (with placeholder 'Enter old password'), 'New Password' (with placeholder 'Enter New Password'), and 'Repeat new password' (with placeholder 'Re-enter new password'). Below the fields are 'Cancel' and 'Update Password' buttons. A sidebar on the left contains icons for home, add, and settings. A help icon is in the bottom right corner.

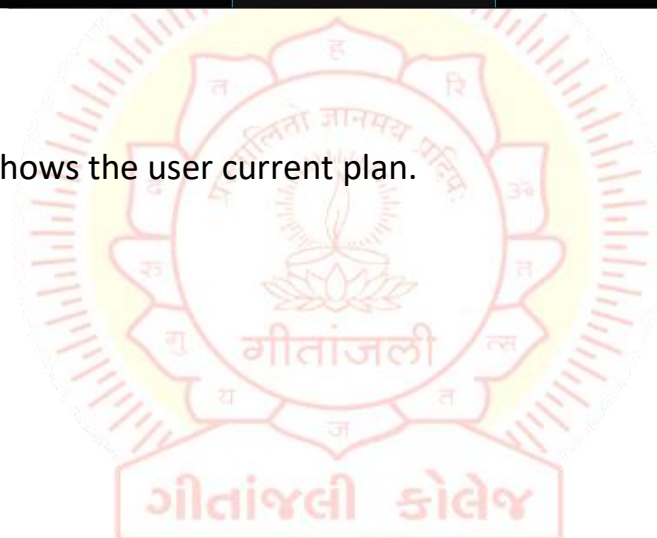
- This page allows user to update their password by entering old and new password.



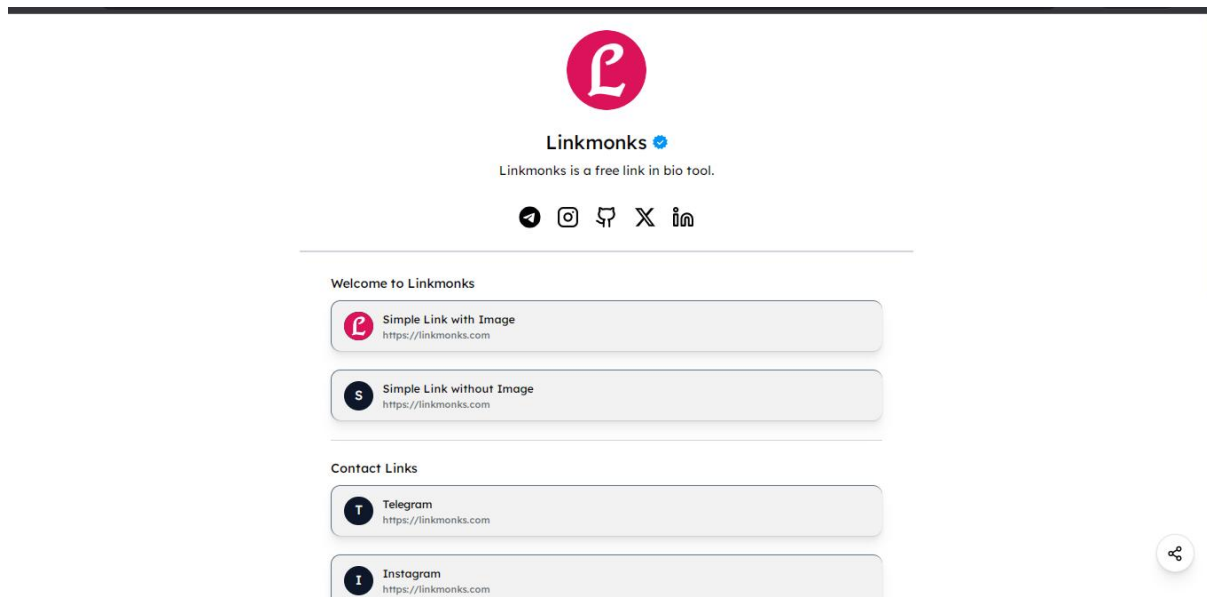
- **Subscription Page:-**



➤ This page shows the user current plan.



- **Single Link Display Page:-**



- This is the main page where user can share with others with their unique url.
- This page also have share feature and more coming soon...



Future Enhancement Of Projects

Enhancing a Link in Bio Tool involves adding new features like password protected blocks, providing email newsletter forms, contact forms, improving existing functionality of stats page.

1.Password Protected Link:

Implement a fast and efficient search system with filters for easy resource discovery.

2. Providing Single Block Stats:

Develop a user-friendly mobile app for on-the-go access to library services.

3.Email Newsletters:

Set up automated reminders for due dates and reservation pickups.

4.Contact Forms:

Integrate barcode scanning for swift check-in and check-out processes.

5.QR Code with company logo:

Include a simple feedback system for user suggestions and opinions.

6.Share single link:

Ensure the system's responsiveness across various devices for a seamless experience.

7.Email Campaigns:

Enable automatic email notifications for overdue items and other important updates.

8.Export Statistics Data:

Streamline data management with the ability to import/export data in bulk.

Limitation Of Project:

The limitations of a Link in Bio project can vary based on the specific implementation and context. Here are some common limitations associated with Linkmonks:

Limited Budget:

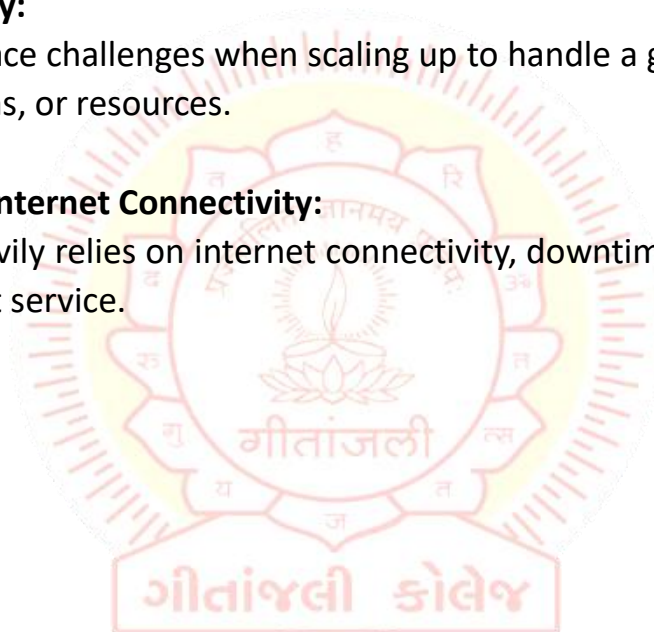
Insufficient funding may restrict the implementation of advanced features and necessary infrastructure.

Limited Scalability:

Some LMS may face challenges when scaling up to handle a growing number of users, transactions, or resources.

Dependency on Internet Connectivity:

If the system heavily relies on internet connectivity, downtime or slow network speed can disrupt service.



Bibliography

- <https://stackoverflow.com/>
- <https://www.javatpoint.com/>
- <https://www.w3schools.com/>

